

Bioptimal Consumables

Bioptimal International Pte. Ltd.

PRUNUS Global R&D Centers





Bioptimal International Pte. Ltd.

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Version: BRH-00-01 Rev.01

Provider of Critical Care Product



30+

Years

100+

Countries

300+

Agents and Distributor

Bioptimal International Pte. Ltd.

Specilized in critical care products over 30 years

Bioptimal International Pte. Ltd. was founded in 1989 and is headquartered in Singapore, one of the most innovative nations in the world.

Being committed to providing world-class medical products, Bioptimal has been building up experience in R&D, as well as production for more than 30 years through repeated experiments and continuous upgrades of manufacturing processes. Consequently, a top quality series of critical care product is forged successfully, including central venous catheters, blood pressure monitoring systems, thermodilution (Swan-Ganz) catheters, embolectomy catheters, bipolar pacing catheters etc. In addition, new products have been constantly under development in order to offer better solutions for health professionals and patients. As the latest member of product portfolio, the silicone foley catheters with patented technologies expand product range into field of urology.



So far, the products of Bioptimal have been highly recognized by both own-brand customers and OEM partners from over 70 countries or regions, especially those markets with high standards, such as, Germany, Sweden, Switzerland, UK, Japan and USA, include other EU territories and Singapore.

Bioptimal would keep up the pace and strive to provide cutting edge medical products to satisfy health professionals and patients.

Business layout & Production base

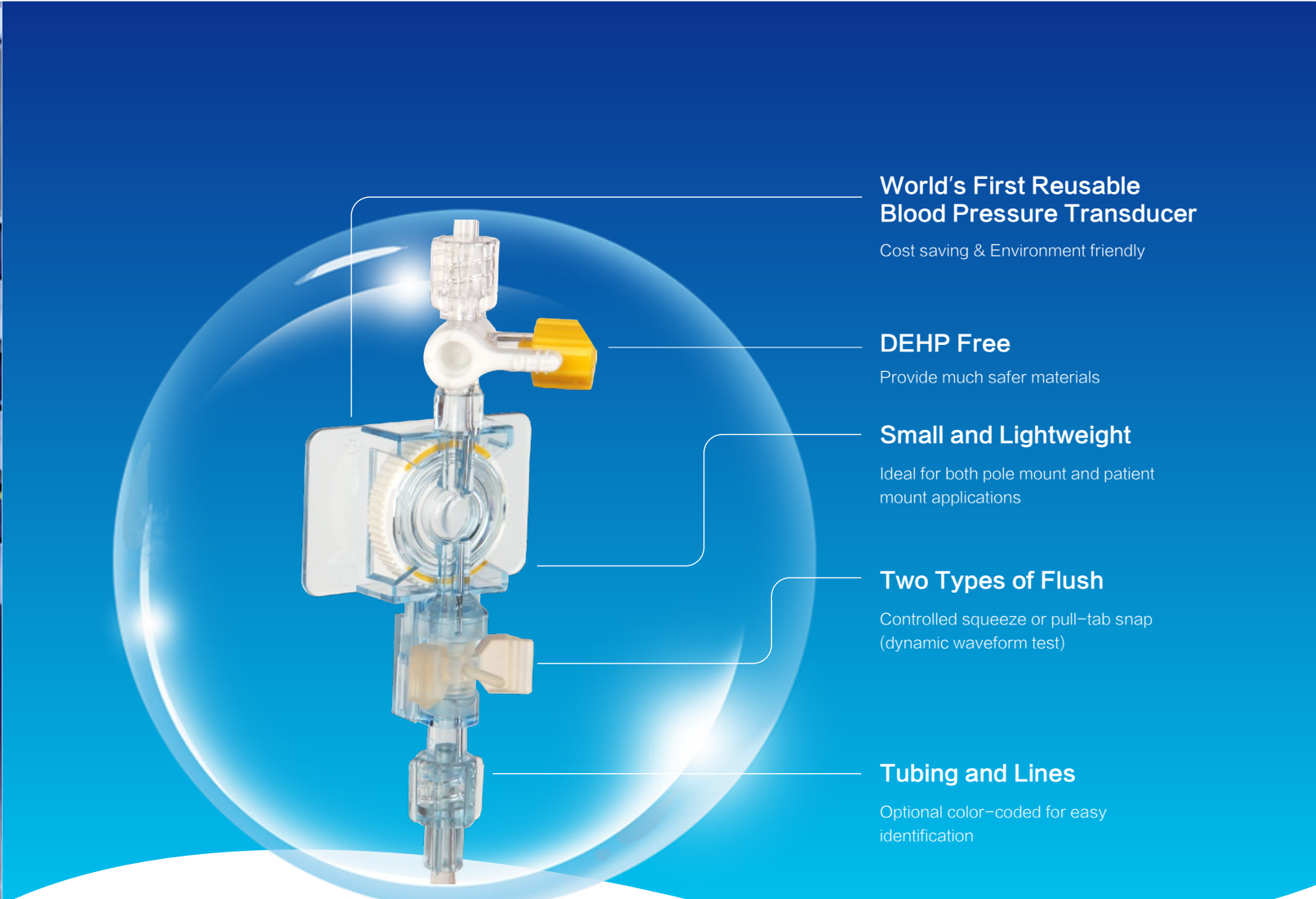
Bioptimal consumables have been registered and certified in more than 50 countries and regions such as the United States, the European Union, Singapore, and China.

The products are sold to more than 100 countries and nearly 300 agents and distributors around the world.



Wei hai Bioptimal consumables production base

- Nearly 5000 m² production workshop and 2800 m² warehouse
- All products have been approved by FDA and MDD 93/42/EEC
- All products have obtained ISO 13485:2016 Quality Management System certification



BIOTRANS™

Semi-Disposable Blood Pressure Transducer

Provide better solutions based on features, quality, operation and environmental friendly.

↓ 30%

It has been calculated that the cost can be reduced over 30% while using semi-disposable blood pressure transducer.

Accurate measurement



Excellent hardware design



Transparent body



Minimizing biological waste



New sensitive sensor chip

Environmental friendly



Patented design



Decrease pollution



Core chip with long life



Easy operation

CONTENTS

All kits include sterilized BIOTRANS disposable Twist–On Dome with integral 3 or 30cc/hr flush device with Fast Flush and pressure relief functions. The table below identifies additional components included in sterile, packaged single–use Pressure Monitoring Kits by catalog number. All luers are supplied with protection vented caps. All required non–vented caps are supplied in a separate pouch within each kit package.

Model No.	Description	Flush Device*	Extension Tube	Stopcock	Distal Tubing
BTR	Standalone DPT Without Flush Device	None	None	None	None
BTR–30**	Standalone DPT With Flush Device (Infant/Neonatal)	30cc/hr	None	None	None
BTR–3	Standalone DPT With Flush Device (Adult)	3cc/hr	None	None	None
BTR 4812–3	Single Monitoring Kit System (Adult)	3cc/hr	48" (122cm)	3–way	12" (30cm)
BTR2 4812–3	Bifurcated Monitoring Kit System (Adult)	3cc/hr	48" (122cm)	3–way	12" (30cm)
BTR3 4812–3	Trifurcated Monitoring Kit System (Adult)	3cc/hr	48" (122cm)	3–way	12" (30cm)
BTR 6012–3	Single Monitoring Kit System with 60" extension tubing (Adult)	3cc/hr	60" (152cm)	3–way	12" (30cm)

* Flush rates indicated are nominal and were obtained with 300mmHg pressure.
Fast Flush rates vary with the type of administration set and length and lumen diameter of the monitoring tubing catheter.
** Kits supplies with 30cc/hr (yellow banded) flush devices are intended for infant and neonatal applications.
Note: Please contract our Bioptimal Sales Representative for custom kits to meet your special requirements.

Resuable BIOTRANS™ Accessories

Model No.	Description
HM650–150	Vertical Mounting Clamp for IV Poles up to 4cm diameter
HM650–150–H	Horizontal Mounting Clamp for IV Poles up to 4cm diameter
HM650–150–R	Rotatable Mounting Clamp for IV Poles up to 4cm diameter in both vertical and horizontal application
BT–201	BIOTRANS II Sensor Base(reusable) with integrated 25cm sensor cable is designed for patient or pole mount application
PMA–1	Replacement Docking Bracket that holds 1 transducer at one time
3–in–1 Backplate	Transducer Holder that holds up to 3 transducer at one time
MAC–XX***	3.05m(10Ft) Monitor Adapter Interface Cables for specific brands of monitors

*** Please contact your Bioptimal representative for assistance.

SPECIFICATIONS

Environmental Performance

- Warehouse Storage Temperature <25℃
- Operating Temperature +15℃ to +40℃(59F to 104F)
- Operating Atmospheric Pressure 10% to 90% non–condensing
- Operating Atmospheric Pressure 425mmHg to 850mmHg

Mechanical Specifications

- Pressure Range –30 to 300mmHg
- Overpressure Withstand –400mmHg to +4000mmHg
- Mounting Any axis

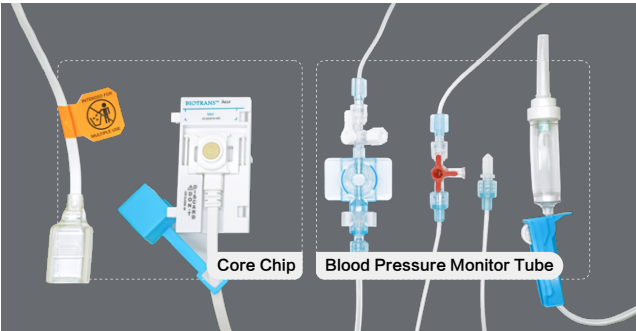
Safety Features

- Risk Current Less than 5uA
- Defibrillation Withstand 5 discharges in 5 minutes @ 360J

Electrical Performance

- Transducer Excitation Voltage 4 to 8 volt RMS
- Transducer Supply Frequency DC to 5000Hz
- Phase Shift <5°
- Transducer Excitation Impedance 300Ω to 400 ohms
- Transducer Signal Impedance 250 to 350 ohms
- Transducer Symmetry +/-5%
- Accuracy Meets AAMI/ ANSI BP22 (1994) Standard
- Sensitivity 5 μ V/V/mmHg
- Zero Drify After 5 min Warm–up <2mmHg in 4hrs
- Zero Shift due to Temperature +/-10mmHg maximum*
- Light** Sensitivity @ 0mmHg <1mmHg
- Mechanical Shock withstands Withstands 3falls from 1 meter

* From 15℃ to 40℃(59F to 104F) using 25℃(77F) as a reference .
** 3,000 foot–candles generated by 3,400K Tungsten source.



Semi–disposable Blood Pressure Transducer

After usage of Biotrans, only the pressure monitoring line is discarded. The core chip is reusable, and can maintain accuracy, which would reduce the pollution of medical waste.



World Class Chip
Provide accurate measurement

Small and Lightweight
Ideal for both pole mount and patient mount applications

Two Types of Flush
Controlled squeeze or pull-tab snap (dynamic waveform test)

Tubing and Lines
Optional color-coded for easy identification

ACCUTRANS™

Pressure Monitoring Systems and Kits

Provide Reliable and Precise Measurements



Excellent hardware design



New sensitive sensor chip

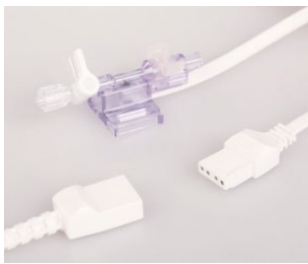


DEHP within limit



Convenient,
easy to set-up and use

ACCUTRANS Accessories



Fully compatible with accessories from the existing Biotrans II Transducer System.

ACCUTRANS Transducer Interface Cable



Bioptimal has the full range of transducer interface cables that is compatible to all commonly-used patient monitors in the market.

Bioptimal makes ordering and stocking your monitoring components easy. Whether you require a standard kit, or prefer a customized version, we are prepared to respond to your critical care monitoring needs.

CONTENTS

All kits include sterilized ACCUTRANS disposable pressure transducer with integral 3 or 30cc/hr flush device with pressure relief functions. The table below identifies additional components included in sterile, packaged single-use Pressure Monitoring Kits by catalog number. All luers are supplied with protection vented caps. All required non-vented caps are supplied in a separate pouch within each kit package.

Model No.	Description	Flush Device*	Extension Tube	Stopcock	Distal Tubing
AT	Standalone DPT Without Flush Device	None	None	None	None
AT-30**	Standalone DPT With Flush Device (Infant/Neonatal)	30cc/hr	None	None	None
AT-3	Standalone DPT With Flush Device (Adult)	3cc/hr	None	None	None
AT 4812-3	Single DPT System (Adult)	3cc/hr	48" (122cm)	3-way	12" (30cm)
AT2 4812-3	Bifurcated DPT System (Adult)	3cc/hr	48" (122cm)	3-way	12" (30cm)
AT3 4812-3	Trifurcated DPT System (Adult)	3cc/hr	48" (122cm)	3-way	12" (30cm)
AT 6012-3	Single DPT System with 60" extension tubing (Adult)	3cc/hr	48" (122cm)	3-way	12" (30cm)

* Flush rates indicated are nominal and were obtained with 300mmHg pressure.
Fast Flush rates vary with the type of administration set and length and lumen diameter of the monitoring tubing catheter.
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HM650-150-R	Rotatable Mounting Clamp for IV Poles up to 4cm diameter in both vertical and horizontal application
PMA-1	Transducer Holder that holds 1 transducer at one time
3-in-1 Backplate	Transducer Holder that holds up to 3 transducer at one time
MAC-XX***	3.05m(10Ft) Monitor Adapter Interface Cables for specific brands of monitors

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SPECIFICATIONS

Environmental Performance

- Storage Temperature<25℃
- Operating Temperature+15℃ to +45℃(59F to 104F)
- Warehouse Storage Humidity0% to 80%
- Operating Atmospheric Pressure400mmHg to 850mmHg

Mechanical Specifications

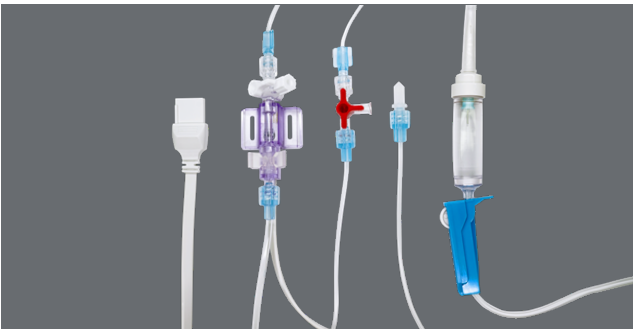
- Pressure Range-30 to 300mmHg
- Overpressure Withstand-400mmHg to +4000mmHg
- MountingAny axis

Safety Features

- Risk CurrentLess than 5mA
- Defibrillation Withstand5 cycles in 5 minutes @ 360J

Electrical Performance

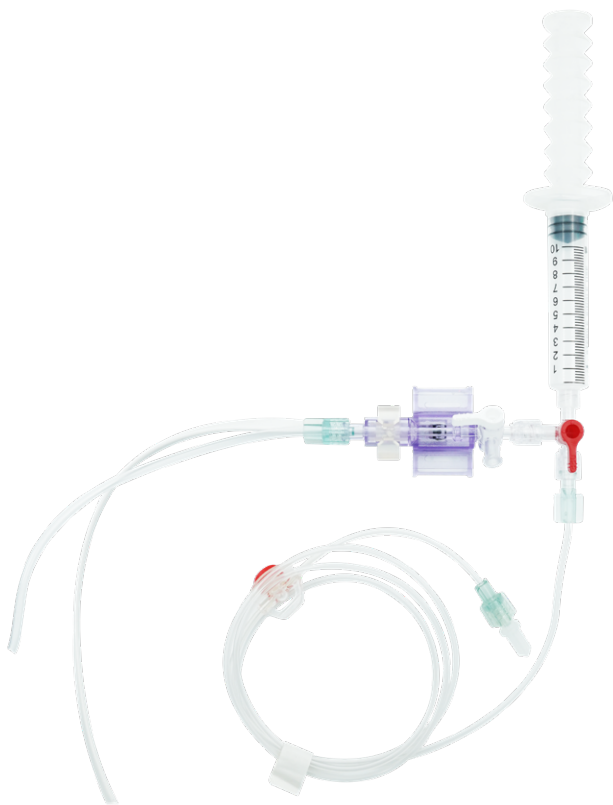
- Transducer Excitation Voltage1 Vrms to 10 Vrms
- Transducer Supply FrequencyDC to 5000Hz
- Phase Shift<5°
- Transducer Excitation Impedance1200Ω to 2200Ω
- Transducer Signal Impedance285Ω to 375Ω
- Transducer Symmetry-5% to +5%
- AccuracyMeets AAMI/ANSI BP22 (1994) Standard
- Sensitivity4.95 μ V/V/mmHg to +5.05 μ V/V/mmHg
- Zero Drify After 5 min Warm-up<1mmHg in 4hrs
- Temperature Sensitivity Coeffivient<0.1%℃
- Zero Offset-20mmHg to +20mmHg
- Light Sensitivity<1mmHg
- Mechanical Shock (2ms sine,3 axis)500g
- Vibration(20Hz to 2000Hz, 1 octave/min,3 axis)20g



Disposable Blood Pressure Transducer

The launch of ACCUTRANS reassures our customers that we are still anticipating to your increasing needs for innovative and ease-of-use Disposable Presure Transducer. By introducing ACCUTRANS to our existing products, we are offering one of the most comprehensive products ranges in the critical care area.

Bioptimal's Needleless Closed Blood Sampling Systems



PRODUCT DESCRIPTION

A closed in-line arterial and venous sampling system designed to reduce unnecessary blood discard and blood spillage hence reduces risk of iatrogenic anaemia and cross contamination.

The pressure monitoring system combines an In-Line Reservoir for waste collection and a stopcock with Marvelous™ handle that enables flushing the line and syringe tip with minimal residual volume.

The In-Line Reservoir eliminates the discarding of waste taken through the set prior to the blood sampling. The entire clearing volume is reinfused to the patient upon completion of the sampling process in a safe manner that minimizes contamination.

This system reduces risk of infection or line contamination when compared to open (or stopcock) sampling devices. Bioptimal's needle-free closed blood sampling system that is customisable for adult and paediatric patients in which protects both clinician and patient.

Closed sampling system

Reduces the risk of nosocomial infection or line sepsis

Needleless sampling system

Reduces the risk of blood spillage during sampling

One-handed operation of reservoir syringe

Makes the system easy to use

Two sampling sites

In one kit (for use in critical care facilities – ICU and OR)

Predetermined drawback volumes

Hence minimising blood waste

Integrated reservoir syringe

Conserves precious blood volume of the patient (small volume blood loss can be critical for neonatal patients)

Safety for Clinicians

Reduces potential exposure from needle stick injury, minimising clinician's exposure to patient blood.

Patient blood pressure monitoring

Maintains continuous patient blood pressure monitoring before and after blood sampling.

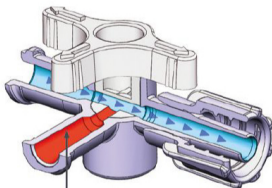
A clear, pure blood sample

Can be easily drawn for accurate lab analysis

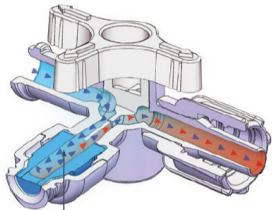
SPECIFICATIONS

Property	Units	Marvelous for monitoring applications (standard flow)	Marvelous for IV application (high flow)
Flow ² rate at "All open" position	[ml/min]	140 ± 5	200 ± 5
Flow ² rate while side port closed	[ml/min]	70 ± 5	105 ± 5
Flow ² rate from side port to male Luer	[ml/min]	135 ± 5	1 ± 5
Side port volume	[ml]	0.2 ± 0.02	0.2+0.02

* 1、 For acceptable quality levels refer to spec # 0022–G
2、 Flow was measured under 75 mmHg pressure/1 meter water pressure



Trapped fluid or air in side port



Constant flushing of the side port

Product Code	Product Description	Integral Flush Device	Extension Tubing	Distal Tubing	Standard Pack Size
AT 8006–3–MV	Single DPT System (Adult) with Closed Arterial Blood Sampling	3cc/hr	80" (203cm)	06" (15cm)	5pcs / box
AT 4812–3–MV2	Single DPT System (Adult) with Dual Closed Arterial Blood Sampling	3cc/hr	48" (122cm)	12" (30cm)	5pcs / box
AT2 7212–3–MV2	Bifurcated DPT System (Adult) with Closed Blood Sampling	3cc/hr	72" (183cm)	12" (30cm)	5pcs / box
AT3 7212–3–MV2	Trifurcated DPT System (Adult) with Closed Blood Sampling	3cc/hr	72" (183cm)	12" (30cm)	5pcs / box
AT 8006–30–MV	Paediatrics–Single DPT System with Closed Arterial Blood Sampling	3cc/hr	80" (203cm)	06" (15cm)	5pcs / box
AT BC–248A	Single DPT System (Adult) with Closed Arterial Blood Sampling	3cc/hr	48" (122cm)	12" (30cm)	5pcs / box
AT2 BC–248A	Bifurcated DPT System (Adult) with Closed Blood Sampling	3cc/hr	48" (122cm)	12" (30cm)	5pcs / box

Note: Customization is available, please contact your local sales representative for more information.

CATRANS

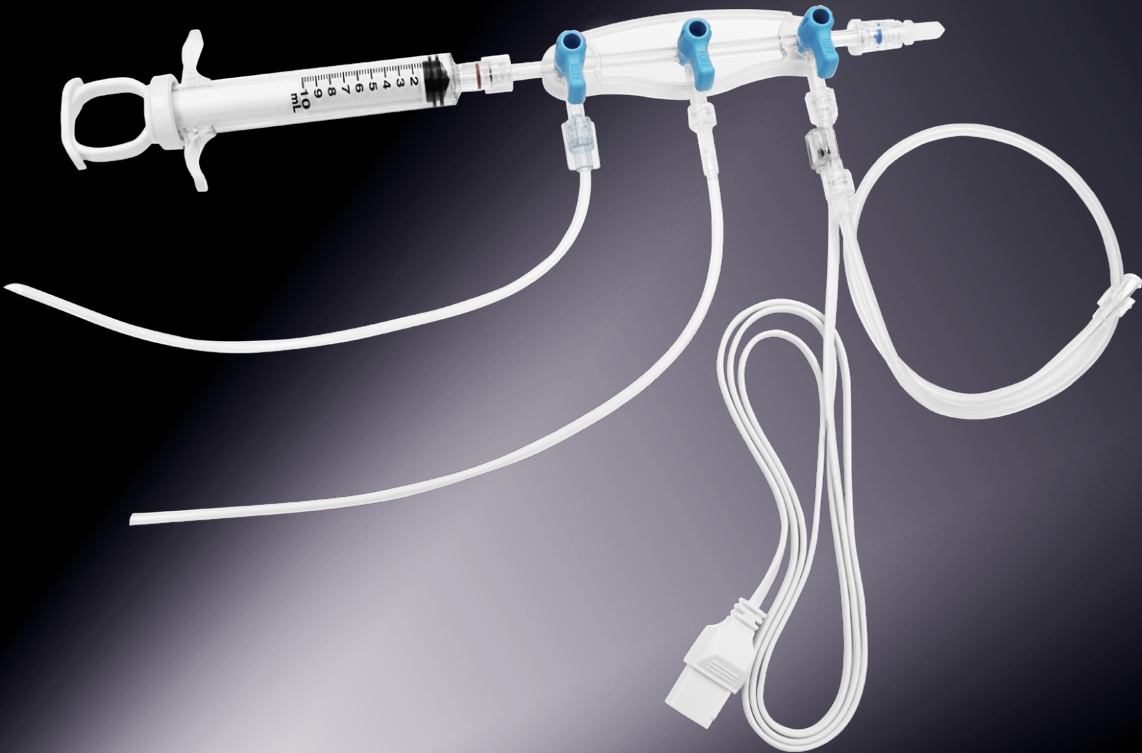
Pressure Monitoring Systems and Kits

More accurate
waveforms

Precise waveform
fidelity

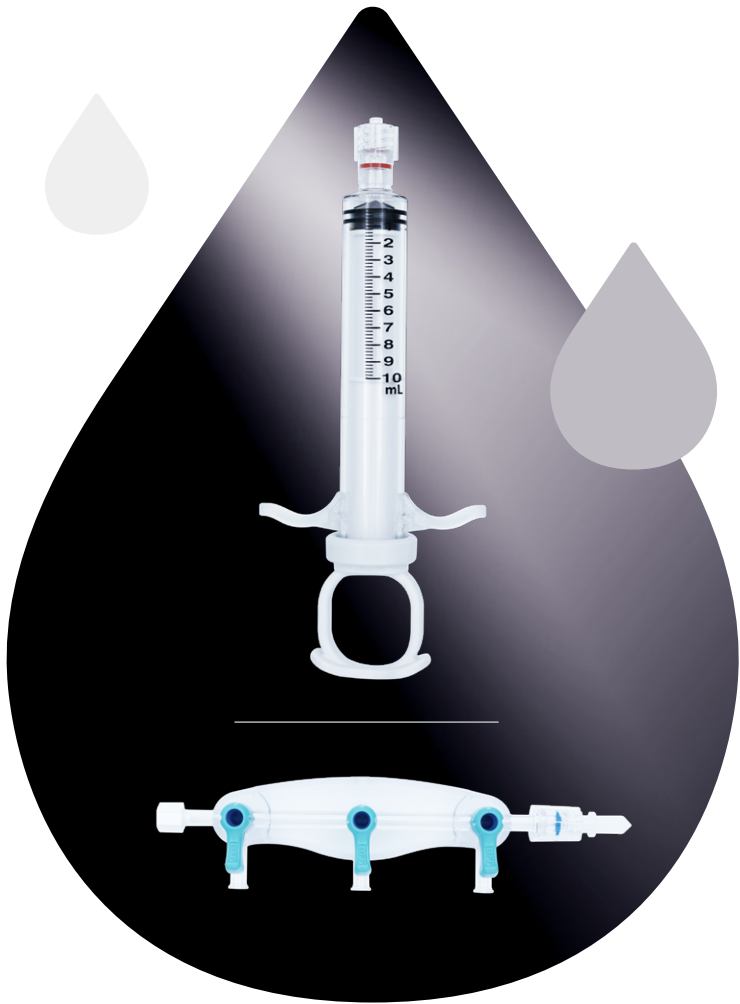
Quick and easy priming,
debubbling and
visualisation

Kits are assembled
to specifications



Note: Customization is available, please contact your local sales representative for more information.

Angiographic Kits



Bioptimal Angiographic Kits consists of one 3–gang manifold, one IV administration set, one contrast media administration set, one pressure tubing and one control syringe.

The male rotating connector at the distal end of the manifold will be connected to PTCA catheters or angiographic catheters that will be inserted into the patient using Seldinger techniques.

INTENDED USE

Bioptimal Angiographic Kits are designed for use in the cath lab during PCI procedures or Angiography to monitor patient's arterial pressure, administer intravenous solution and contrast media to the patient.

COMPONENTS

Multi–Gang Manifold	IV Administration Set
Control Syringe	IV Tubing
High Pressure tubing	Pressure Tubing

Model Code	Main Component	Notes
Prime Angiographic Kit	3–Gang Manifold IV Administration Set Contrast Media Administration Set Pressure Tubing Control Syringe	High pressure tubing (optional)
Prime Angiographic Kit II	3–Gang Manifold IV Administration Set Contrast Media Administration Set Pressure Tubing	High pressure tubing (optional)
3–GMF, RH(OFF)–550 psi	3–Gang Manifold	550 psi
3–GMF, RH(OFF)–500 psi	3–Gang Manifold	500 psi
3–GMF, RH(ON)–550 psi	3–Gang Manifold	550 psi
3–GMF, RH(ON)–500 psi	3–Gang Manifold	500 psi
2–GMF, RH(OFF)–500 psi	2–Gang Manifold	500 psi
2–GMF, RH(ON)–500 psi	2–Gang Manifold	500 psi

Note: Customization is available, please contact your local sales representative for more information.



Central Venous Catheter and Catheterization Kit

Your reliable choice for deep vein catheterization



Safety & stable

Special formulated biocompatible polyurethane material and soft tip



Smooth operation

Special designed guidewire with a soft flexible tip, catheter tip with multi-hole design



Multi-choice

Basic CVC Kit and CVC Procedure Tray for choice. Customization is available

Catheter Marking

Radiopaque material with clear, definite marking, ensures correct placement of the catheter.

Suture Wing & Clamp Fastener

Keep catheter in place thereby reducing the risk of mechanical phlebitis and irritation.

Polyurethane Material

Special formulated biocompatible polyurethane material, provides strength during insertion, yet soften in body.

Soft Tip

Superior geometric soft tip design makes insertion safe and easy.

COMPONENTS

Polyurethane Catheter

"J" Tip Guidewire Complete with Dispenser & Advancer

Introducer Needle(Straight Needle or Y Valve Needle)

Tegaderm Transparent Dressing(Optional)

Needle Sponge

Vessel Dilator

5cc Syringe with Luer Slip

Scalpel #11, short

Suture Wing & Clamp Fastener

Extension Line Clamp

Male Luer Lock Injection Site

Customization is available, please contact your local sales representative for more information.

SPECIFICATIONS

Model No.	Catheter Size	Gauge Size	Length(cm)							Spring Guidewire Size			Dilator	Introducer Needle	
			6	8	10	13	16	20	30	0.018"	0.025"	0.035"		18Gx21/2"	20Gx11/2"
Single Lumen															
CV-401-16	4F	18G					✓				✓		5F		✓
CV-401-20	4F	18G						✓			✓		5F		✓
CV-501-16	5F	16G					✓					✓	6F	✓	
CV-501-20	5F	16G						✓				✓	6F	✓	
CV-701-16	7F	14G					✓					✓	7F	✓	
CV-701-20	7F	14G						✓				✓	7F	✓	
Double Lumen															
CV-402-06	4F	22/22G	✓							✓			5F		✓
CV-402-10	4F	22/22G			✓					✓			5F		✓
CV-402-16	4F	22/22G					✓			✓			5F		✓
CV-502-10	5F	18/18G			✓						✓		6F		✓
CV-502-16	5F	18/18G					✓				✓		6F		✓
CV-502-20	5F	18/18G						✓			✓		6F		✓
CV-702-16	7F	16/16G					✓					✓	8F	✓	
CV-702-20	7F	16/16G						✓				✓	8F	✓	
CV-702-30	7F	16/16G							✓			✓	8F	✓	
CV-802-16	8.5F	13/13G					✓					✓	10F	✓	
CV-802-20	8.5F	13/13G						✓				✓	10F	✓	
CV-802-30	8.5F	13/13G							✓			✓	10F	✓	
Triple Lumen															
CV-503-08	5.5F	20/22/22G		✓						✓			6F		✓
CV-503-13	5.5F	20/22/22G				✓				✓			6F		✓
CV-503-30	5.5F	20/22/22G							✓	✓			6F		✓
CV-703-10	7F	16/18/18G			✓							✓	8F	✓	
CV-703-16	7F	16/18/18G					✓					✓	8F	✓	
CV-703-20	7F	16/18/18G						✓				✓	8F	✓	
CV-703-30	7F	16/18/18G							✓			✓	8F	✓	
CV-713-10	7F	14/18/18G			✓							✓	8F	✓	
CV-713-16	7F	14/18/18G					✓					✓	8F	✓	
CV-713-20	7F	14/18/18G						✓				✓	8F	✓	
CV-713-30	7F	14/18/18G							✓			✓	8F	✓	
CV-803-16	8.5F	14/18/18G										✓	10F	✓	
CV-803-20	8.5F	14/18/18G										✓	10F	✓	
CV-803-30	8.5F	14/18/18G										✓	10F	✓	
Quad Lumen															
CV-804-16	8.5F	14/16/18/18G					✓					✓	10F		
CV-804-20	8.5F	14/16/18/18G						✓				✓	10F		
CV-804-30	8.5F	14/16/18/18G							✓			✓	10F		

OPTIONAL FEATURES

Family/Configuration	Description of configuration	Remark
CV-XXX-XX	CVC with Spring Guidewire and Straight Needle	NA
CVN-XXX-XX	CVC with Kink-Resistance Guidewire and Straight Needle	N-Refer to Kink-Resistance Guidewire
CVNR-XXX-XX	CVC with Kink-Resistance Guidewire and Raulerson Syringe	R-Refer to Raulerson Syringe
CV-XXX-XX Y	CVC with Spring Guidewire and Y-Valve Needle	Y-Refer to Y-Valve Needle
CV-XXX-XX T	CVC Kit with Procedure Tray	T-Refer to Procedure Tray
CVR-XXX-XX	CVC with Raulerson Syringe	R-Refer to Raulerson Syringe

All complete CVC kit includes Extension Line Clamp, Male Luer Lock Injection Site, 1 pcs of Luer Slip Syringe, 1 pcs of #11 Short Scalpel, 1 pcs of Suture Wing with Clamp Fastener & 1 pcs of Needle Sponge.

QUALITY COMPONENTS FOR OPTIMAL PERFORMANCE



Tegaderm Transparent Dressing (Optional)

Transparent window provides visibility , stabilized catheter and protects from outside contamination.

Introducer Needle

Available in straight or Y needle. Y valve needle designed to prevent back-flow of blood and air embolism.

Advancer

Single hand operated advancer helps to straighten the "J" Tip spring guidewire and advancing it to the proper position.

Needle Sponge

Specialized sponge for holding used needle preventing the potential of needle piercing.

Guidewire

Specially designed guidewire with a soft flexible tip to prevent vessel perforation(vascular trauma), yet provides good torque to ensure firm insertion. Markings on guidewire provide a good insertion indication.

Contents of the CVC Procedure Tray

CVC Kit	Sponge Swabs	Fenestrated Drape
Needle 25G x 5/8"	Suture: Braided Silk 3-0 with Straight Needle	5cc syringe with luer slip
Paper Towel 17" x 22"	CVC Tray-Medication Cup	Blue Wrap (70cm x 70cm)
Gauze White	Needle 22G x 11/2"	Tegaderm transparent dressing (Optional)

The CVC Procedure Tray is optional and can be purchased together with the standard central venous catheter kit as a complete set. Please approach our sales representatives for further enquires.



Thermodilution Catheter

Gold standard for diagnosing pulmonary hypertension



Safety & stable

Special formulated biocompatible polyurethane material



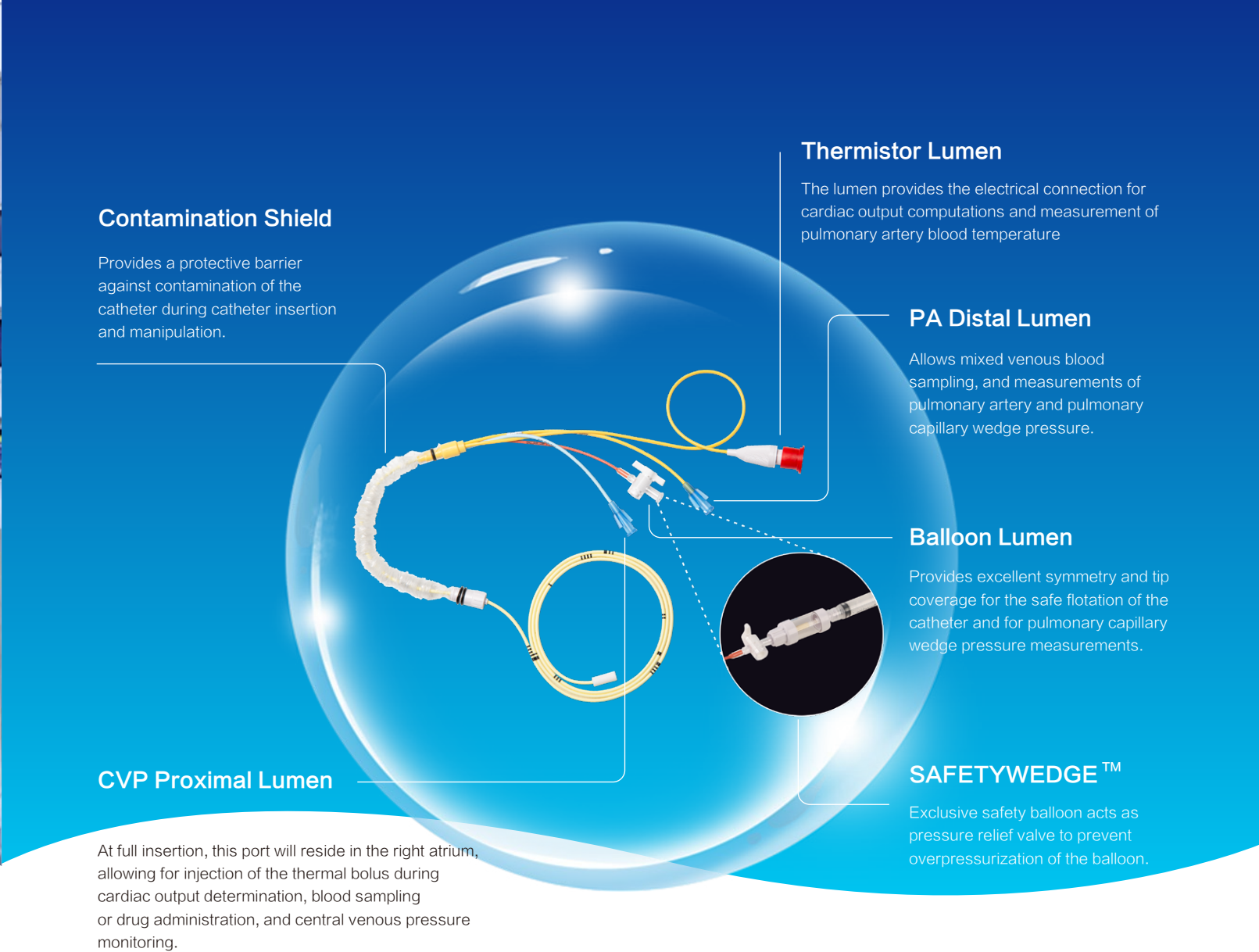
Smooth operation

Special designed contamination Shield provides a protective barrier against contamination of the catheter during catheter insertion and manipulation



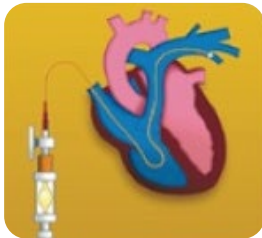
Multi-choice

Basic package and TD/PA KIT for choice, customization is available



ELIMINATING THE RISK OF PULMONARY ARTERY RUPTURE

Patent design——SAFETYWEDGE, UNITED STATES patent No.:5,007,919



Catheter tip migration into small PA branch, SAFETYWEDGE device activated upon attempt to inflate distal balloon.



Correct catheter position, with SAFETYWEDGE device in READY mode.

SPECIFICATIONS

Model No.	Material	Catheter Size	Lumen	Length	Balloon Volume(cc)	Thermistor Resistance(ohms)	Recommended Introducer Size
Thermodilution Catheter							
TD1504N	PVC	5F	4	90CM	0.75	14K	6F
TD2504N	PU	5F	4	90CM	0.75	14K	6F
TD1604N	PVC	6F	4	110CM	1.00	14K	7F
TD2604N	PU	6F	4	110CM	1.00	14K	7F
TD1704N	PVC	7F	4	110CM	1.50	14K	8F
TD2704N	PU	7F	4	110CM	1.50	14K	8F
TD1755N	PVC	7.5F	5	110CM	1.50	14K	8.5F
TD2755N	PU	7.5F	5	110CM	1.50	14K	8.5F
PA Monitoring Catheter							
TD1502N	PVC	5Fr	2	90CM	0.75	N/A	6F
TD2502N	PU	5Fr	2	90CM	0.75	N/A	6F
TD1602N	PVC	6Fr	2	110CM	1.00	N/A	7F
TD2602N	PU	6Fr	2	110CM	1.00	N/A	7F
TD1702N	PVC	7Fr	2	110CM	1.50	N/A	8F
TD2702N	PU	7Fr	2	110CM	1.50	N/A	8F
TD1603N	PVC	6Fr	3	110CM	1.00	N/A	7F
TD2603N	PU	6Fr	3	110CM	1.00	N/A	7F
TD1703N	PVC	7Fr	3	110CM	1.50	N/A	8F
TD2703N	PU	7Fr	3	110CM	1.50	N/A	8F

Radiopacity: Sufficiently opaque to appear visible under conventional fluoroscope illumination while in vivo.

Special Features

Optional Features	Suffix	Description
SAFETYWEDGE™	D	Catheter with SAFETYWEDGE™
Contamination Sleeve	X	Catheter without Contamination Sleeve
Non–Coated Tubing	N	Catheter with no coating on tubing
Stiff body tubing	F	Catheter with stiffer tubing

Note: Customization is available, please contact your local sales representative for more information.

The TD/PA KIT contains all the necessary supplies used with the SAFETYWEDGE™ thermodilution catheter. Value added benefits:

- Elimination of the sheath–catheter size compatatible issue
 - Elimination of nursing cover as the tray can be opened in a sterile area
 - All components matched by BIOPTIMAL for optimum compatibility
- Ease of use
 - Easier inventory control
 - Convenience

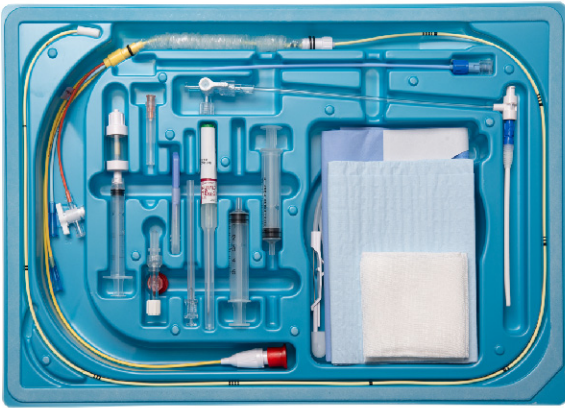
CONTENTS OF THE TD/PA KIT

Features high quality products at lower costs, saves time and reduces supply inventories.

One Thermodilution Catheter/PA Monitoring Catheter	Gauze Swab	Contamination Shield (Optional)	Venous Introducer with Dilator
SAFETYWEDGE Balloon Device (Optional)	Needle 25G	Guidewire (0.035")	Paper Towel 17 x 22"
Needle 18G x 2 1/2"	Syringe Luer Slip 5cc	Needle 22G	Drape Minor Proc. Fen 22 x 22"
Needle 18G x 2 1/2" OTN Catheter	Scalpel #11 Short	Suture. Silk	

ADVANTAGE OF PU CATHETER

- Polyurethane performs better against thrombosis, eliminating the need for Heparin coating.
 - Polyurethane is tough, biocompatible, and hemocompatible. Polyurethane also outperforms many other materials in flexibility, tear resistance and abrasion resistance. Polyurethane is stiff when insert the catheter however after it goes into vein and contacts blood it will become soft which can reduce the risk of damage and injury to the vein wall.
 - The use of Polyurethane in Bioptimal’s thermodilution catheters eliminates the problems associated with other materials such as PVC, where the dangers of leachable plasticizers has become a concern.
 - Polyurethane is radiopaque material which can be detected clearly to ensure correct placement of the catheter.

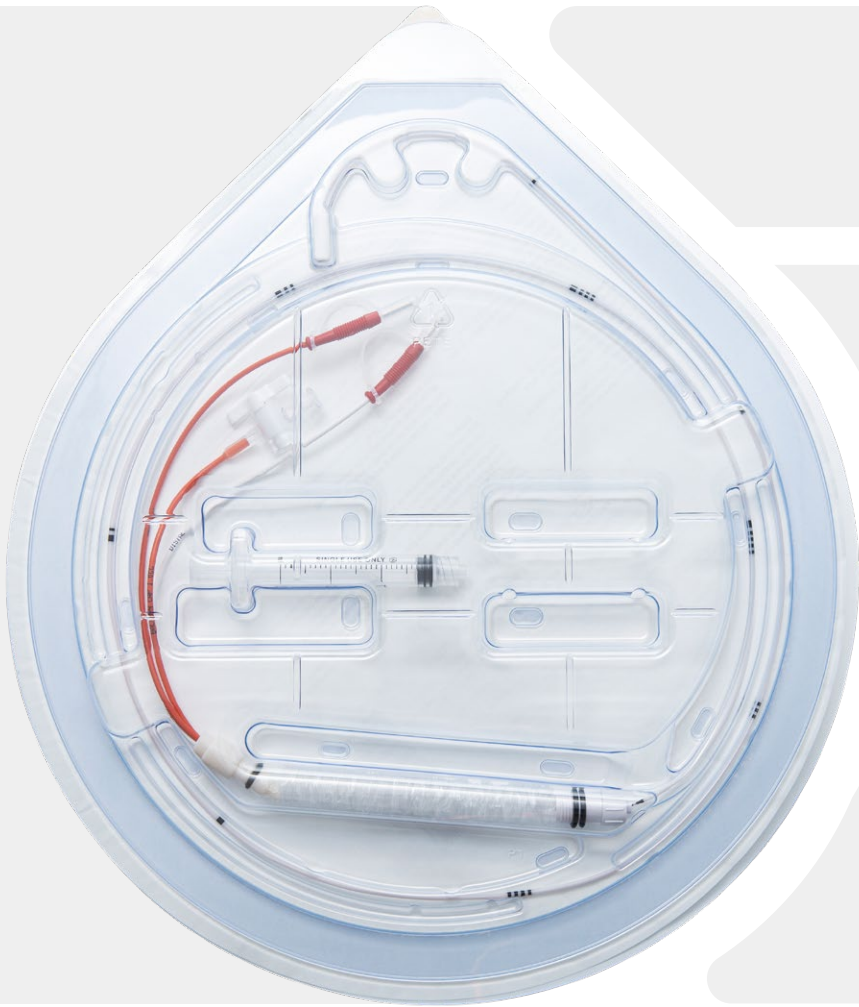


Total Package, Total Solution

Enjoy the benefits of the SAFETYWEDGE™ thermodilution catheter in a convenient, fully accessorised all-in one tray.

Bipolar Pacing Catheter

Secure, accurate device for critical procedures



J-Tip

Right Heart Curve towards the heart helps the catheter reach the right ventricle quickly

Safe

Contamination Shield—meets clinical requirements of pollution protection

Accurate

Tubing is marked in 10cm increments and radiopaque polyurethane

Optional safety ensurance

An optional inflation balloon—located at the Distal tip—can be inflated to enable safe flotation of the catheter to the right ventricle



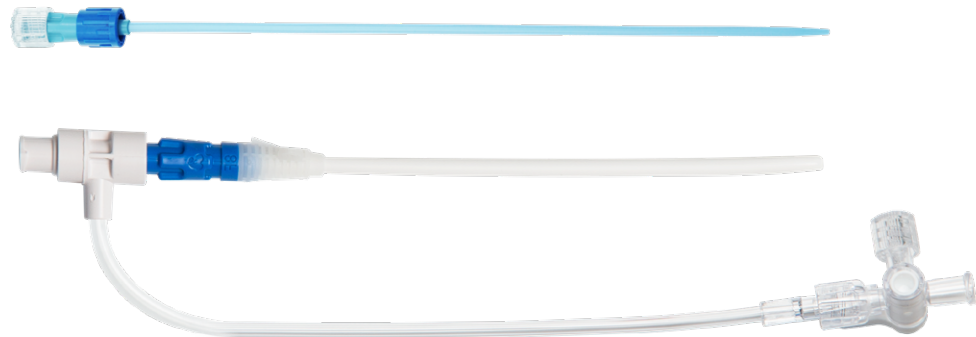
SPECIFICATIONS

Model No.	Catheter Size	Length	Inflated Balloon Capacity	Maximum Inflation Diameter	Electrode Width	Electrode Spacing	Contamination Shield	Recommended Introducer Size
With Balloon								
BP2502-10	5F	110cm	1cc	9mm	3mm	10mm	NA	6F
BP2502CS-10	5F	110cm	1cc	9mm	3mm	10mm	110cm	6F
Without Balloon								
BPX2502-10	5F	110cm	NA	NA	3mm	10mm	NA	6F
BPX2502CS-10	5F	110cm	NA	NA	3mm	10mm	110cm	6F

Note: Customization is available, please contact your local sales representative for more information.



Vascular Introducer Kit



PRODUCT DESCRIPTION

Vascular Introducer Kit is a basic customized kit that comprises of an introducer with or without a vessel dilator, a guidewire, together with some other components mainly to assist in the catheterization procedures.

PRINCIPLE OF OPERATION

Vascular Introducer Kit is intended to assist in the introduction of temporary pacing leads, balloon catheters, closed end catheter or angiographic catheters into the vasculature.

SPECIFICATIONS

Model No.	Size (Fr.)	Length (cm)	Guidewire
CPIK6F	6F	10cm	0.035" (0.89mm) × 60cm
CPIK7F	7F	10cm	0.035" (0.89mm) × 60cm
CPIK8F	8F	10cm	0.035" (0.89mm) × 60cm
CPIK8.5F	8.5F	10cm	0.035" (0.89mm) × 60cm

Optional Accessories

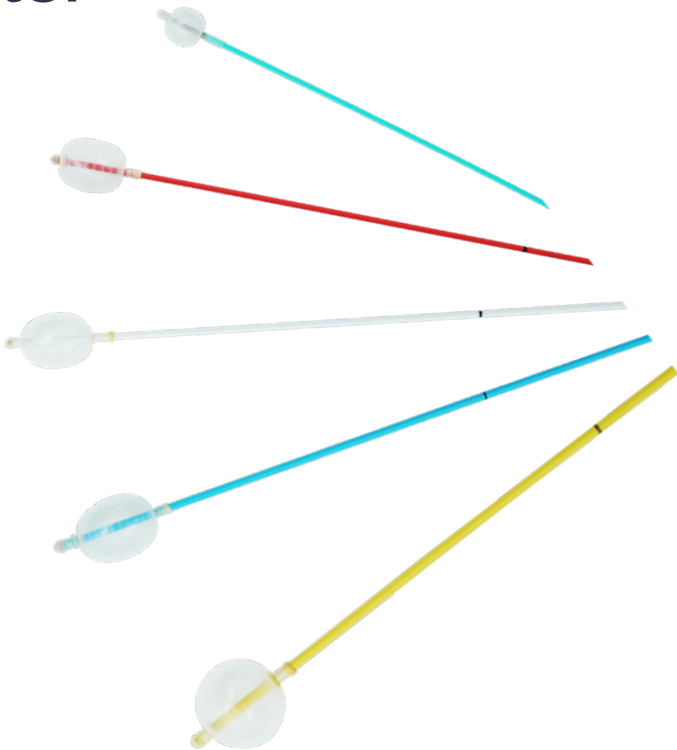
18G x 2 1/2" Needle	Syringe Luer Slip 5mL/10mL	OTN Catheter	Guide Wire (0.035"/0.89mm– J60cm)
Scalpel # 11–Short	Gauze Swab 7.5x7.5x8PLY	3–0 Sutures	Paper Towel 17" x 22"
Needle 25G or 22G	Drape Minor Proc. Fen 22x22"	Syringe 1.5cc	

Note: Customization is available, please contact your local sales representative for more information.

Embolectomy Catheter

BENEFITS

- Bioptimal Embolectomy Catheter offers simple, rapid removal of fresh, soft emboli from the peripheral arterial system.
- Proprietary “low-profile” winding technique for balloon attachment insures balloon symmetry, providing uniform contact with vessel walls.
- Soft, rounded tip promotes easy arterial insertion for reduced trauma, arteriopuncture and plaque avulsion.
- Flexible polyvinylchloride body with “workhorse” latex balloon.
- Winged hub for easy catheter manipulation during insertion and attachment of balloon inflation syringe.
- Available in 4 lengths and 6 diameters, ranging from 2 French to 7 French.
- Size and balloon inflation capacity printed on proximal end for convenient and rapid indentification.



Model No.	Catheter Size	Length	Colour Code	Maximum Liquid Volume	Maximum CO ₂ Volume	Inflated Balloon Diameter	Maximupu Pull Force Inflated Balloon (lb)
EL 3260	2F	60cm	Purple	0.08cc	0.2cc	4mm	0.5
EL 3280	2F	80cm	Purple	0.08cc	0.2cc	4mm	0.5
EL 3340	3F	40cm	Green	0.20cc	0.6cc	6mm	0.7
EL 3380	3F	80cm	Green	0.20cc	0.6cc	6mm	0.7
EL 1440	4F	40cm	Red	0.75cc	1.7cc	9mm	1.5
EL 1480	4F	80cm	Red	0.75cc	1.7cc	9mm	1.5
EL 14100	4F	100cm	Red	0.75cc	1.7cc	9mm	1.5
EL 1540	5F	40cm	White	1.50cc	3.0cc	11mm	2.0
EL 1580	5F	80cm	White	1.50cc	3.0cc	11mm	2.0
EL 1640	6F	40cm	Blue	2.00cc	4.5cc	13mm	2.5
EL 1680	6F	80cm	Blue	2.00cc	4.5cc	13mm	2.5
EL 1740	7F	40cm	Yellow	2.50cc	5.0cc	14mm	3.5
EL 1780	7F	80cm	Yellow	2.50cc	5.0cc	14mm	3.5

Note: Customization is available, please contact your local sales representative for more information.

Silicone Foley Catheter

Provide a better solution for female catheter intubation to prevent infection



Innovative Design

The specially designed advancer is used to accurately locate the female urethral opening, fully isolate the catheter to avoid infection, and insert the urethral catheter into bladder smoothly.



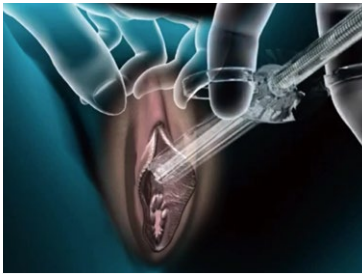
Benefit

To reduce the risk of microbe taken into urethra and bladder while inserting a catheter. To reduce injury to urethral mucosa caused by catheterization.

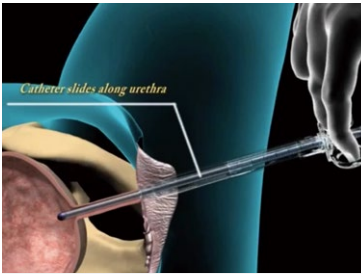


Application

Applicable to female patients in all clinical departments.



Circumferential positioning, precise guidance, complete isolation



Stably low insertion, uniform intubation

Less pain Less infection



PATENT
[POSITIONING]
DESIGN

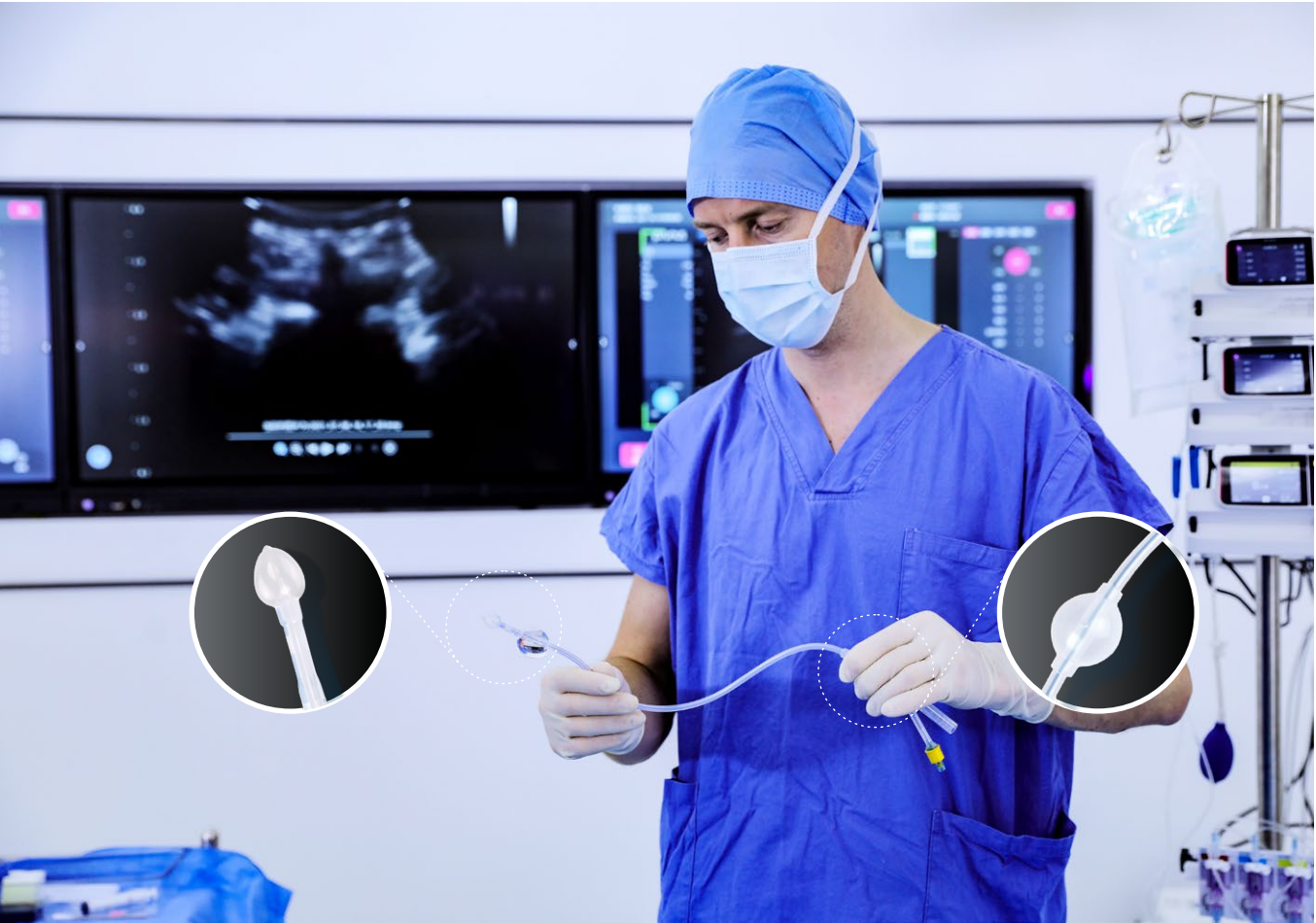
SPECIFICATIONS

Model No.	Description	Size (FR/Ch)	Balloon Size (mL)	Length (mm)
FC214W I	2-way	14	5-15	410
FC214W II	2-way	14	20-30	410
FC216W I	2-way	16	5-15	410
FC216W II	2-way	16	20-30	410
FC218W I	2-way	18	5-15	410
FC218W II	2-way	18	20-30	410
FC220W I	2-way	20	5-15	410
FC220W II	2-way	20	20-30	410
FC314W I	3-way	14	5-15	410
FC314W II	3-way	14	20-30	410
FC316W I	3-way	16	5-15	410
FC316W II	3-way	16	20-30	410
FC318W I	3-way	18	5-15	410
FC318W II	3-way	18	20-30	410
FC320W I	3-way	20	5-15	410
FC320W II	3-way	20	20-30	410

Note: Customization is available, please contact your local sales representative for more information.

Silicone Foley Catheter

Top penetration balloon dilated foley catheter
Subvert traditional operation of catheterization



Innovative Design

When the intubation is blocked, the top segment will go into the urethra's narrow part first, and then the top balloon will be filled with gas thus expanding the urethra radially, which enables the catheter to pass smoothly.



Benefit

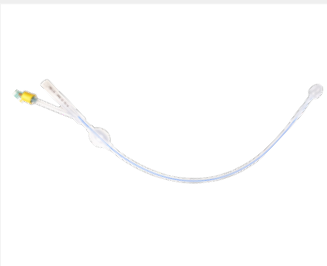
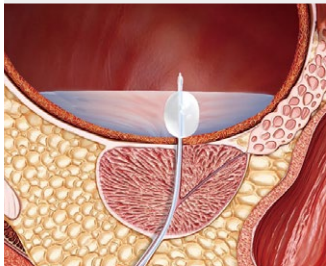
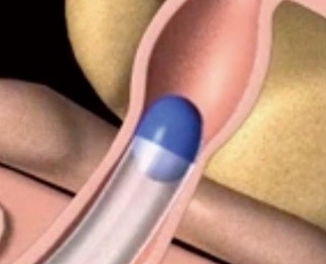
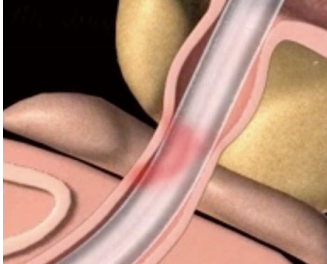
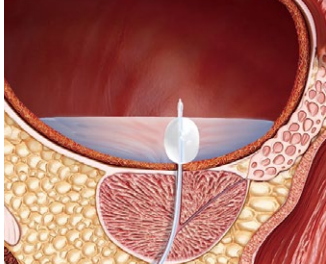
Can minimize the urethra and mucosa damage in the process of intubation, alleviate patients' pain, lower urinary tract infections and complications.



Application

Suitable for physiological (urethral bending, etc.), pathological (top R hypertrophy, etc.) or psychological factors (such as tension) in urine into narrow indwelling urine of patients.

Silicone Foley Catheter–Soft Tip Type **VS** Traditional Foley Catheter

Items	When meet urethral stricture	Injury and pain in the narrow place	Finished intubation
			
Silicone Foley Catheter–Soft Tip Type	Balloon section of catheter will pass through the urethral structure	Reduce urethral injury and pain	Less pain, less infection
			
Traditional Foley Catheter	Difficult to pass through the urethral structure	Urethra and mucous membrane are easily damaged, and patients have obvious pain	Increase infection and scar stenosis is formed

SPECIFICATIONS

Model No.	Description	Size (FR/Ch)	Balloon Size (mL)	Length (mm)
FC314S I	3-way-2-funnel	14	5-15	470
FC314S II	3-way-2-funnel	14	20-30	470
FC316S I	3-way-2-funnel	16	5-15	470
FC316S II	3-way-2-funnel	16	20-30	470
FC318S I	3-way-2-funnel	18	5-15	470
FC318S II	3-way-2-funnel	18	20-30	470
FC320S I	3-way-2-funnel	20	5-15	470
FC320S II	3-way-2-funnel	20	20-30	470

Note: Customization is available, please contact your local sales representative for more information.